

# HAT-P-19b

R-1161

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-19\_231102 · created 2026-07-31T02:26:33+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460250.8475 +/- 0.0049 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.179 +/- 0.053                 |
| Transit depth (Rp/Rs)^2                           | 3.21 +/- 1.9 [%]                |
| Orbital Inclination (inc)                         | 89.65 +/- 1.57                  |
| Airmass coefficient 1 (a1)                        | 1.23 +/- 0.59                   |
| Airmass coefficient 2 (a2)                        | -0.15 +/- 0.32                  |
| Scatter in the residuals of the lightcurve fit is | 45.61 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.76                            |
| Optimal Annulus                                   | 6.75                            |
| Transit Duration (day)                            | 0.0837 +/- 0.0087               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.179 $\pm$ 0.053 vs 0.1418 (deviation 0.61 $\sigma$ ) |
| Duration fit vs published            | 0.0837 d vs 0.1196 d (deviation 3.58 $\sigma$ )        |
| Mid-time O-C                         | 12.0 min                                               |
| Depth SNR                            | 2.9                                                    |
| Residual scatter                     | 45.61 %                                                |

### Light curve

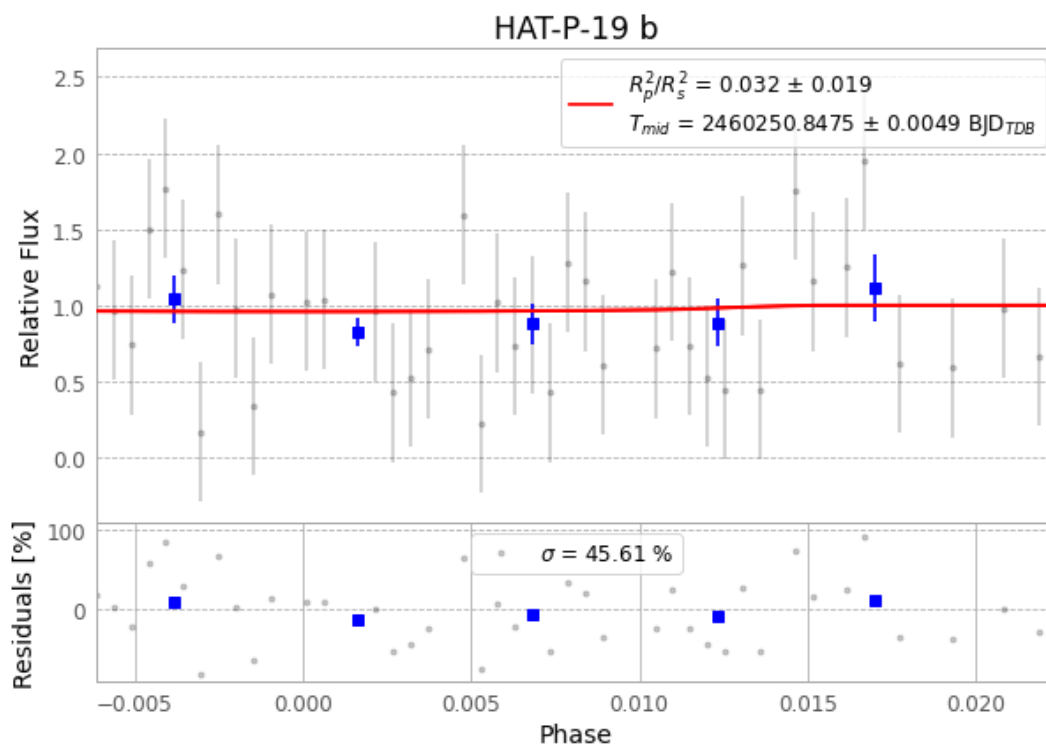


Figure 1 — FinalLightCurve\_HAT-P-19 b\_2023-11-02.png (SHA-256 ab59a007039177f7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [213, 435], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 6c46f64bc02bbff3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

59 FITS frames (39 MB), HATP-19231102073315.FITS → HATP-19231102102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231102.log (6d3eef3a48db...), FinalLightCurve\_HAT-P-19 b\_2023-11-02.png (ab59a0070391...), FinalLightCurve\_HAT-P-19 b\_2023-11-02.csv (c926222b7a3b...)

## Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:26Z.